Qn			Answers	Marks
5	(a)		Each transition between electron energy levels involves the emission or absorption of one photon. The energy of the photon corresponds to the energy difference between the two levels.	B1
			Line spectra consist of discrete lines, showing that atoms only emit or absorb photons with specific energies / frequencies. This provides evidence that electrons in isolated atoms occupy discrete energy levels.	B1

	(b)	**(i)**	Difference in energy of first excited state and the ground state corresponds to the photon absorbed with the lowest energy (longest wavelength). energy of photon $= \frac{hc}{\lambda}$ energy of photon $= \frac{(6.63\times10^{-34})(3.00\times10^{8})}{500\times10^{-9}} = 3.978\times10^{-19}$ J	M1
			Hence, energy of first excited state is… $3.978\times10^{-19}\ \text{J} = E_2 - E_1$ $E_2 = -7.60\times10^{-19} + 3.978\times10^{-19}$ $= -3.62\times10^{-19}$ J	A1
	(b)	**(ii)**	The reason why the emission spectrum has one more line is due to the transition from E_3 to E_2. $E_3 - E_2 = \frac{hc}{\lambda_{1\to3}} - \frac{hc}{\lambda_{1\to2}} = hc\left(\frac{1}{300\times10^{-9}} - \frac{1}{500\times10^{-9}}\right)$ energy of photon $= \frac{hc}{\lambda_{3\to2}}$ $hc\left(\frac{1}{300\times10^{-9}} - \frac{1}{500\times10^{-9}}\right) = \frac{hc}{\lambda_{3\to2}}$	M1
			$\lambda_{3\to2} = 7.50\ \times\ 10^{-7}$ m	A1